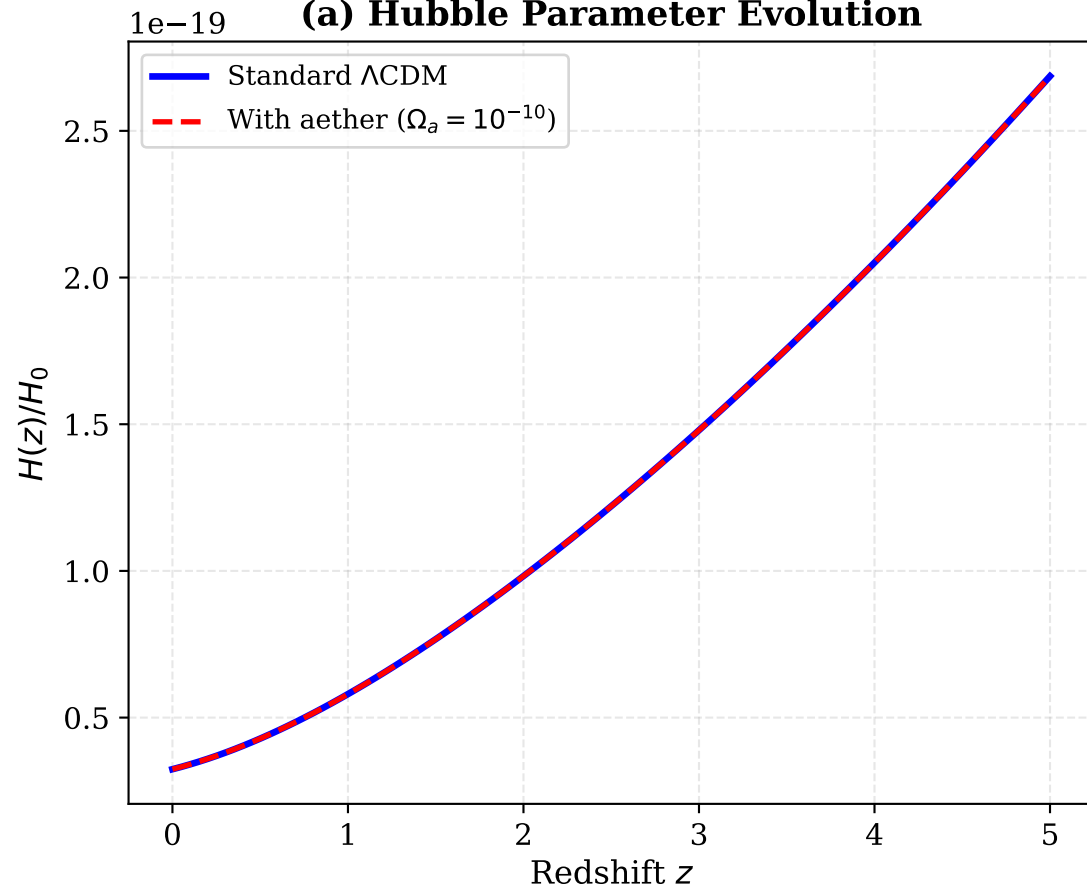
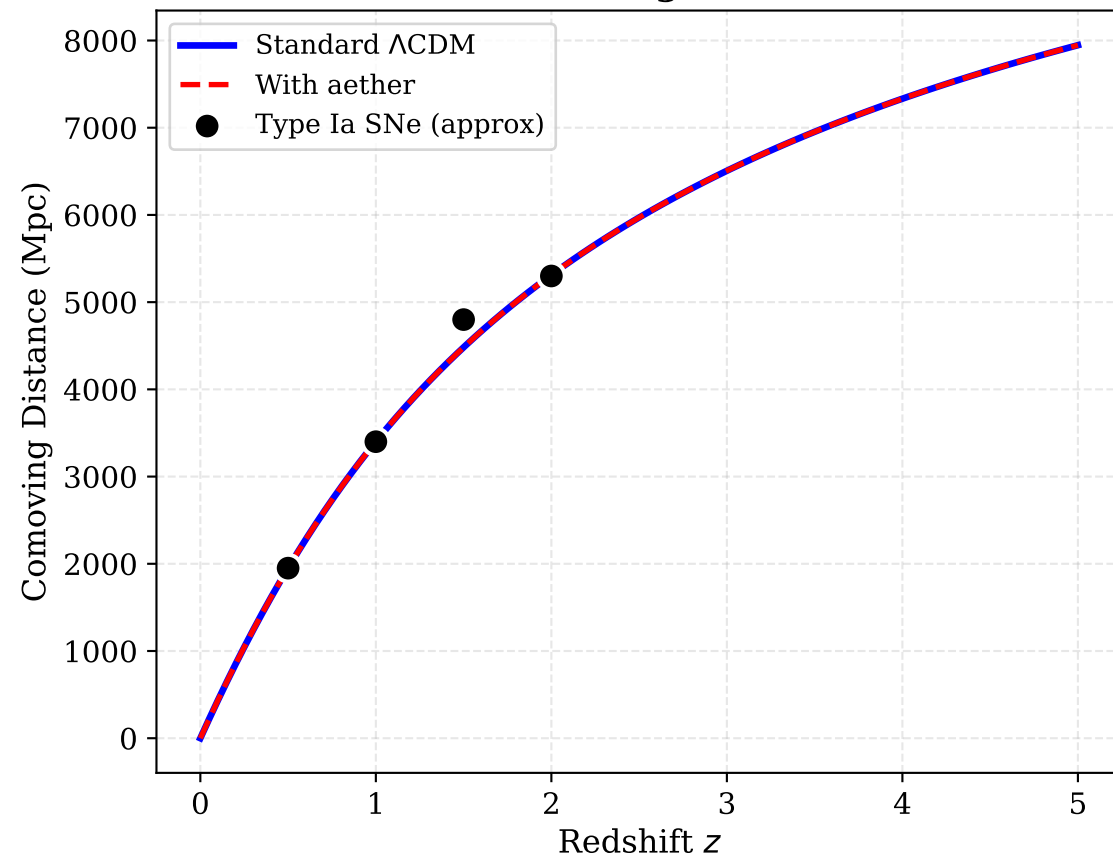
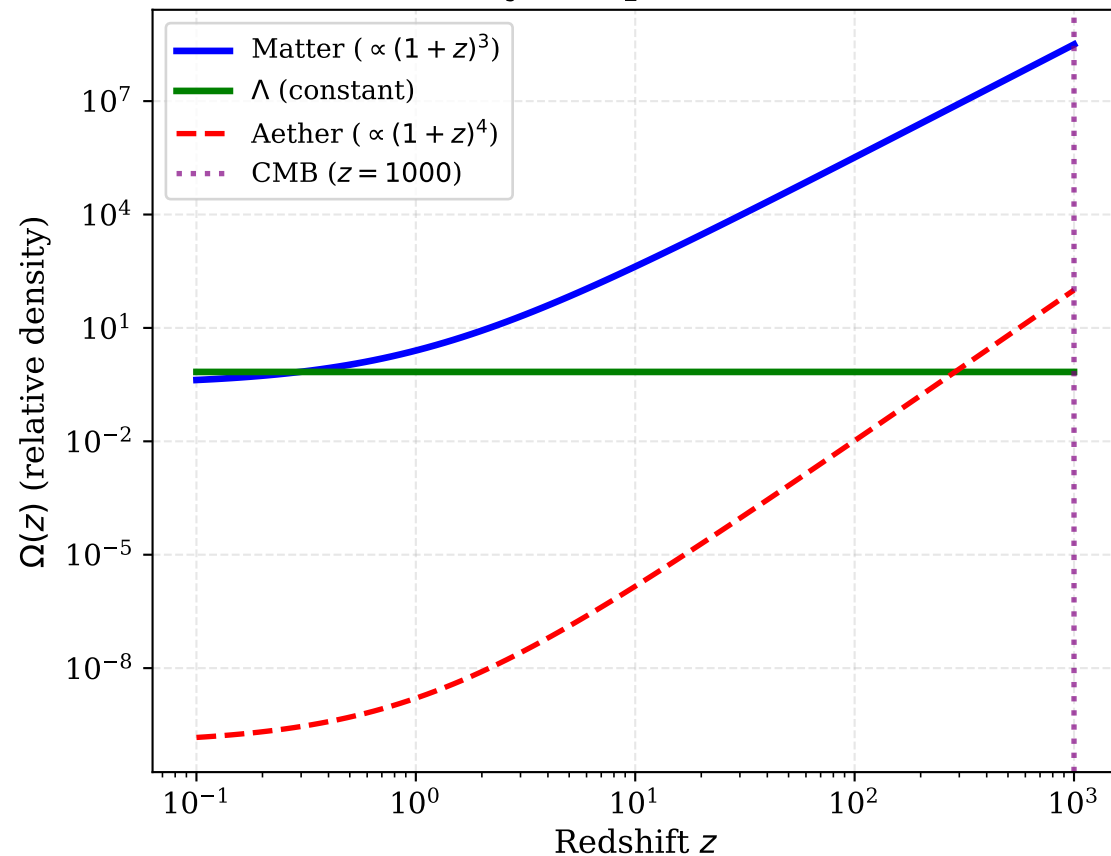


(a) Hubble Parameter Evolution**(b) Comoving Distance****(c) Density Component Evolution****CMB Epoch Constraints ($z = 1000$)****Aether contribution at CMB:**

$$\Omega_{\text{aether}}(z = 1000)/\Omega_m(z = 1000) \approx 10^{-6}$$

Equation of state:

- Matter: $w = 0$ (non-relativistic)
- Λ : $w = -1$ (cosmological constant)
- Aether: $w = -1/3$ (radiation-like)

Power spectrum impact:

Aether contribution $< 0.01\%$ at CMB epoch, consistent with Planck 2018 constraints on additional radiation-like components.

Future tests:

- High- z supernovae ($z > 5$)
- CMB B-mode polarization
- 21-cm cosmology (epoch of reionization)
- Gravitational wave standard sirens